EFU CONSTRUCTION LICENCE v1.0

Experimental EFU-based Construction & Building Materials Framework

0. Purpose and Scope

Experimental EFU-based licensing and measurement framework for construction (concrete, buildings, infrastructure, roads).

NOT a building permit, structural safety, or land use authorization replacement, but an open research protocol measuring construction's physical footprint in human-scale EFU metrics (concrete, steel, land use, energy). Covers residential, commercial, civil engineering projects.

I. Core Concepts

1. **EFU-Construction (EFU-CONST)**
   EFU framework application for building activities:
   - **EFU-M_CONST**: concrete, cement, steel, aggregates
   - **EFU-E_CONST**: diesel (machinery), electricity (mixers, cranes)
   - **EFU-L_CONST**: land sealing + soil compaction
   - **EFU-W_CONST**: water (concrete mixing, dust control)
   - **EFU-EM_CONST**: construction dust + truck emissions
2. **Construction Sovereignty Gap (SS_CONST)**
   SS_CONST = TEFU_CONST_total – Regional Construction Carrying Capacity (RCCC_CONST)
   Measures if building activity stays within regional land/material regeneration capacity.
3. **MROI_CONST**
   MROI_CONST = THI_CONST / TEFU_CONST
   THI_CONST = housing supply + infrastructure utility + durability/lifespan.

II. EFU Construction Licence – Research Status

Documents EFU-audited construction firms, concrete plants, infrastructure projects.
Does NOT replace planning permissions, OSHA safety, or environmental impact assessments.
Enables high-rise vs. modular housing benchmarking.

III. Metrics and Thresholds

1. EFU Components (per m² floor area)

EFU-M_CONST (Materials)
EFU-M_CONST = concrete_steel_t / 300kg/person/year / m²
Example: Standard office = 0.8t/m² → 0.0027 EFU/m²

EFU-E_CONST (Energy)
EFU-E_CONST = energy_MWh / 12.88 / m²

EFU-L_CONST (Land)
EFU-L_CONST = sealed_m² / 100m²/person/year / m²_floor

TEFU_CONST = 1.6×EFU-M + 1.2×EFU-E + 1.8×EFU-L + 1.1×EFU-W + 1.3×EFU-EM

2. Regional Construction Carrying Capacity (RCCC_CONST)

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RCCC_CONST = Sustainable_floor_m²M / Population / 20m²/person/year
Example: Germany = 50M m² / 84M / 20m² = 0.03 EFU_CONST/person/year
```

IV. Licence Categories (EFU-CONST Labels)

Level	SS_CONST	MROI_CONST	Label	Interpretation
1	≤ 0	≥1e-2	**EFU-CONST SOVEREIGN**	Modular/renovation
2	0 < ≤20k EFU	≥5e-3	**EFU-CONST BALANCED**	Mixed new/retrofit
3	20k < ≤100k EFU	≥1e-3	**EFU-CONST DEFICIT**	Land constraint
4	>100k EFU	<1e-3	**EFU-CONST CRITICAL**	Material exhaustion

V. Regional RCCC_CONST Calibrations

Country	Floor Area (M m²/year)	Population (M)	RCCC_CONST (EFU/person/year)
GERMANY	50	84	0.03
CHINA	2,000	1,410	0.07
USA	1,200	340	0.18
INDIA	300	1,400	0.0002
EU	400	448	0.045

VI. Example: German Construction Sector

Input (annual sector):

- Floor area: 50M m²
- Concrete: 40M tonnes

- Steel: 4M tonnes
- Diesel: 5B liters (17B MWh)
- Workers: 2M staff
- Region: Germany (RCCC_CONST = 0.03 EFU/person/year)

Calculation:

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EFU-M_CONST = 44M t/300/50M m² = 0.0029 EFU/m²
EFU-E_CONST = 17B/12.88/50M = 0.026 EFU/m²

TEFU_CONST_total ≈ 2,800,000 EFU/year
Allocated = 2M × 0.03 = 60,000 EFU/year
SS_CONST = 2.8M – 60k = +2,740,000 EFU/year (Level 4: CRITICAL)

Result: EFU-CONST CRITICAL
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Sub-sector Comparison: Timber housing = 0.4 EFU/m² (Level 2: BALANCED)

VII. Implementation Notes

1. **EU Taxonomy**: TEFU_CONST for sustainable building certification
2. **CBAM extension**: EFU-M_CONST basis for concrete import duties
3. **Urban land tax**: EFU-L_CONST metric for land sealing charges
4. **Data sources**: Tier 1 = concrete delivery notes; Tier 2 = Eurostat construction stats